Logarithmic functions

1 Consider the function $y = \log_2 x$.

a Complete the table of values.

x	$\frac{1}{2}$	1	2	4	16
y					

b Sketch a graph of the function.

c State the equation of the vertical asymptote.

d State whether the following elements are key features of the graph of $y = \log_2 x$:

i A y-intercept

ii A horizontal asymptote

iii An x-intercept

iv A lower limiting value

v An upper limiting value

2 Sketch the graph of $y = \log_5 x$.

3 Consider the function $y = \log_3 x$.

a Complete the table of values.

x	$\frac{1}{243}$	$\frac{1}{3}$	1	3	9	81
y						

b Is $y = \log_3 x$ an increasing or decreasing function?

c Describe the behavior of $y = \log_4 x$ as x approaches 0.

d State the value of y when $x = 0$.

Inverses of logarithmic functions

4 Consider the function $y = \log_2 x$.

a Complete the table of values for $y = \log_2 x$:

x	1	2	4	8
y				

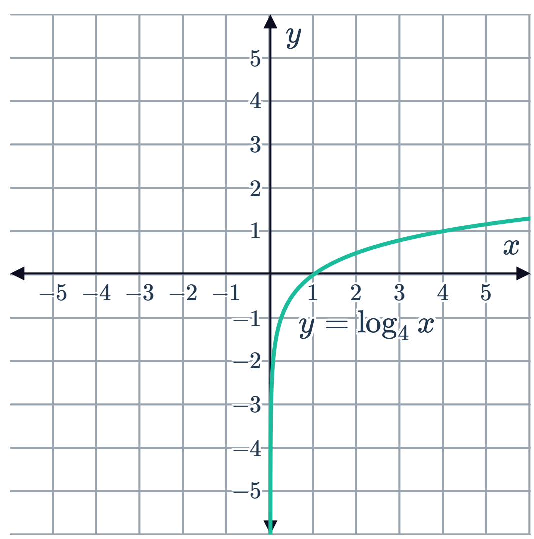
b Complete the table of values for the inverse function of $y = \log_2 x$:

x				
y				

c Sketch the graph of $y = \log_2 x$ and its inverse function on the same coordinate plane.

d State the equation of the inverse function of $y = \log_2 x$.

5 Consider the graph of $y = \log_4 x$:
Sketch the graph of the inverse of $y = \log_4 x$.



6 For each of the following functions:

i Graph $f(x)$, the line $y = x$ and the inverse of $f(x)$ on the same coordinate plane.

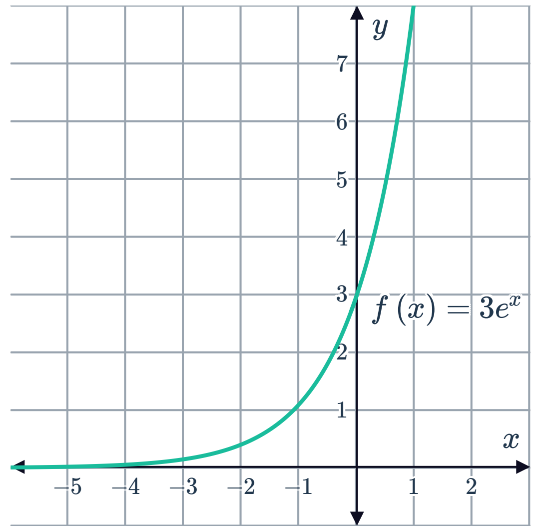
ii State the equation of the inverse function.

a $f(x) = 4^x$ **b** $f(x) = \left(\frac{1}{4}\right)^x$ **c** $f(x) = \left(\frac{1}{5}\right)^x$

Natural logarithmic functions

7 Describe the logarithmic function known as $\ln x$ in terms of its base.

8 Consider the graph of $f(x) = 3e^{x}$:



a Sketch the graph of the inverse of $f(x)$.

b State the equation of the inverse function.

9 Consider the function $f(x) = 2e^{x} - 5$ for all $x \in \mathbb{R}$.

a State the equation of the inverse function.

b Using a graphing calculator, state the coordinates of the points at which the graphs of $y = f(x)$ and $y = f^{-1}(x)$ intersect. Round all values to three decimal places.

Transformations of logarithmic functions

10 If the function $f(x) = \log_3 x$ is translated 5 units to the right, state the equation of the resulting function.

11 Sketch the graph of $y = \log_3 x$ after it has been translated 4 units down.

12 For each of the following functions:

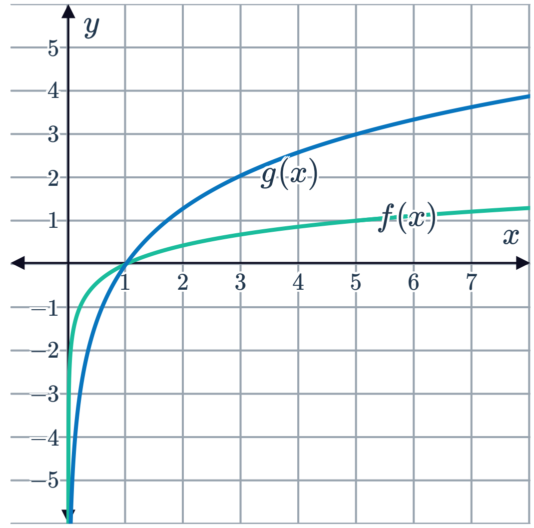
i State the equation of the function after it has been translated.

ii Sketch the translated graph.

a $y = \log_5 x$ translated downwards by 2 units.

b $y = \log_3 (-x)$ translated upwards by 2 units.

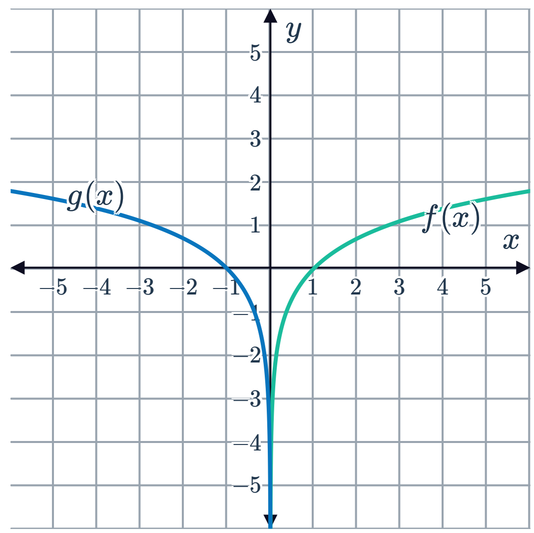
13 Consider the graphs of $f(x) = \log_5 x$ and $g(x)$:
Find the equation of $g(x)$.



14 Consider the graphs of $f(x) = \ln x$ and $g(x)$:

a Describe the transformation that has been performed on the graph of $f(x) = \ln x$ to obtain $g(x)$.

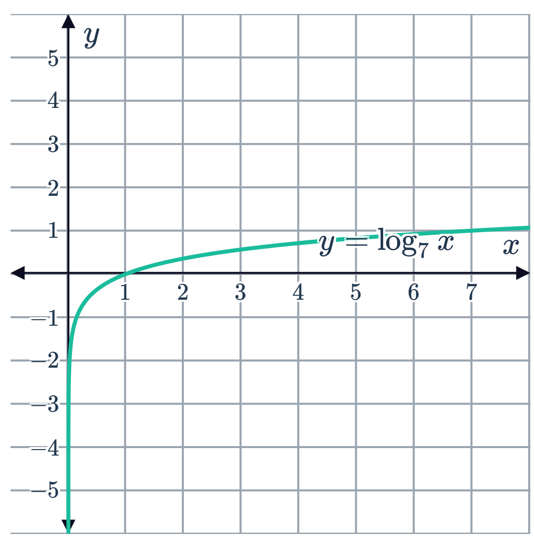
b State the equation for $g(x)$.



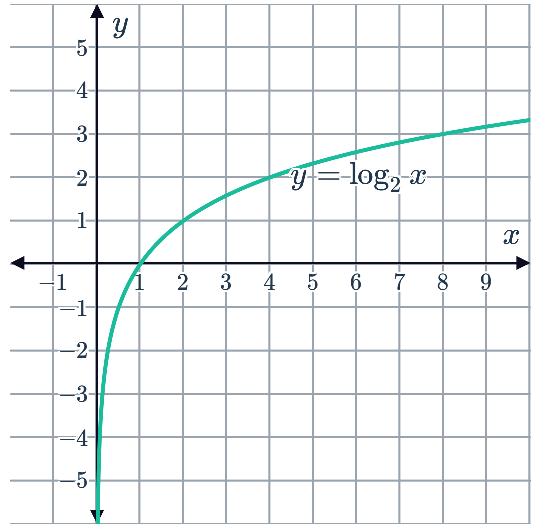
15 Consider the graph of $y = \log_7 x$:

a Describe the transformation needed to obtain the graph of $y = -3\log_7 x$ from $y = \log_7 x$.

b Sketch the graph of $y = -3\log_7 x$ on the same coordinate plane as $y = \log_7 x$.



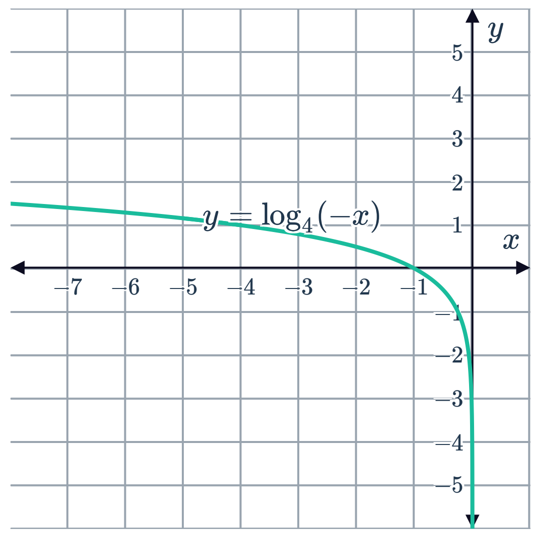
16 Given the graph of $y = \log_2 x$, sketch the graph of the following functions:



a $y = \dfrac{1}{3} \log_2 x$

b $y = -\dfrac{1}{2} \log_2 x$

17 Consider the graph of the function $y = \log_4 (-x)$:



a Complete the table of values for $y = \log_4 (-x)$:

x	-4	-1	$-\dfrac{1}{4}$
$\log_4 (-x)$			

The graph of $y = \log_4 (2 - x)$ can be obtained from the original graph by transforming it.

b Complete the table of values for $y = \log_4 (2 - x)$:

x	-2	1	$\dfrac{7}{4}$
$\log_4 (2 - x)$			

c Describe the transformation from $y = \log_4 (-x)$ to $y = \log_4 (2 - x)$.

18 Consider the function $f(x) = \log_3 (x - 4)$.

a State the equation of the vertical asymptote of $f(x)$.

b State the coordinates of the x-intercept of the function.

c Determine the exact value of $f(7)$.

d Sketch a graph of $f(x) = \log_3 (x - 4)$.

19 Consider the function $f(x) = \log_3(2 - x)$.

a State the equation of the vertical asymptote of $f(x)$.

b State the coordinates of the x-intercept of the function.

c Determine the exact value of $f(-7)$.

d Sketch a graph of $f(x) = \log_3(2 - x)$.

20 The graph of $y = \log_4 x$ has a vertical asymptote at $x = 0$. By considering the transformations that have taken place, state the equation of the vertical asymptote of the following functions:

a $y = 2\log_4 x - 4$

b $y = 2\log_4 x$

c $y = \log_4(x - 5)$

d $y = -\log_4 x$

e $y = \log_4(x + 3) - 2$

f $y = 3\log_4 x + 2$

21 For each of the following functions:

i Solve for the x-intercept.

ii State the equation of the vertical asymptote.

iii Sketch the graph of the function.

a $f(x) = 4\log_2(x - 7)$

b $f(x) = -\log_4(x + 4)$

c $f(x) = \log_2(x - 1) - 4$

d $y = -2\log_3(x + 8)$

e $y = -\log_4(x + 9) + 1$

f $y = -4\log_3(x - 1) - 4$

g $y = \log_2(-x - 1) - 1$

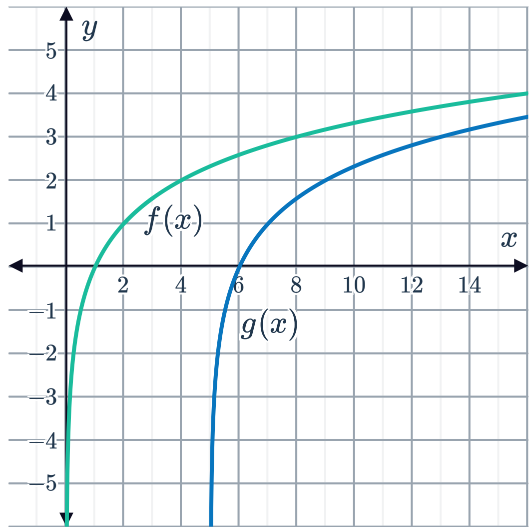
h $y = -\log_2(-x + 2) + 2$

Domain and range of logarithmic functions

22 Consider the graphs of the functions $f(x) = \log_2 x$ and $g(x) = \log_2 (x - 5)$:

a Describe the transformation from $f(x)$ to $g(x)$.

b State whether the domain and range of $f(x)$ change in its transformation to $g(x)$.



23 For each of the following functions:

i State the equation of the inverse function.

ii State the domain of the inverse function using interval notation.

iii State the range of the inverse function using interval notation.

a $f(x) = \log_e 5x$ for all $x > 0$

b $f(x) = \log_e (4x - 3)$ for all $x > \dfrac{3}{4}$

c $f(x) = 4\log_e (2x - 5)$ for all $x > \dfrac{5}{2}$

d $f(x) = 6\log_e 5x - 3$ for $x > 0$

e $f(x) = 6\log_e (x - 4) - 3$ for $x > 4$

f $f(x) = e^{x+3}$ for all $x \in \mathbb{R}$

g $f(x) = e^x - 3$ for all $x \in \mathbb{R}$

h $f(x) = 5 - e^{-x}$ for all $x \in \mathbb{R}$

i $f(x) = 2e^{x+3}$ for all $x \in \mathbb{R}$

j $f(x) = 3e^{5x} + 4$ for all $x \in \mathbb{R}$

24 State the domain of the following functions as an inequality:

a $y = \log 3x$

b $y = \log(-7x)$

c $y = \log\left(\dfrac{1}{2}x\right)$

d $y = \log\left(-\dfrac{1}{4}x\right)$

e $y = \log(3x + 9)$

f $y = \log_3 (x + 5) - 4$

g $y = \log_2 \left(x^2 - 10x + 21\right)$

h $y = \log\left(x^3 - 16x\right)$

i $y = 3\log_{\frac{1}{4}} (x + 5)$

j $y = -4\log_2 (3 - x)$

k $y = \ln\left(x^2 + 9\right)$

l $y = \ln\left(-x^2 + 25\right)$